

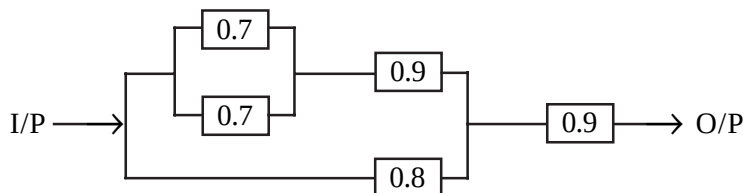
Roll No

AU/ME-801(C) (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Reliability and Maintenance
(Elective - III)
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define Availability of a mechanical system.
b) Compare reliability and quality.
2. What is meant by MTBF and MTTR? Explain the difference between hazard rate function and reliability function in reliability.
3. Define the following terms:
 - i) Hazard Rate
 - ii) Gamma Distribution
 - iii) Unreliability
 - iv) Maintainability Availability
4. Calculate the reliability of the system shown in figure.



5. What is productive maintenance? Differentiate between shutdown maintenance and scheduled maintenance.

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6. Explain the Markov analysis. Explain Tie Set and Cut Set method of reliability evaluation.

OR

Explain about FMECA. Write its elements, applications and benefits.

7. What is Failure rate and how it is used in the calculation of Reliability?

8. Write short technical notes on following (any two):

- a) Visual monitoring
- b) MTTR
- c) K-out-of-m Systems
- d) Risk Evaluation and Priorities
